# Mitigating Muscle Fatigue in Upper-Limb Prosthesis Users Through Exoskeletal Weight Compensation

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***Abstract*— Prosthesis users often experience muscle fatigue and reduced control due to the weight of the device, contributing to high abandonment rates. This study investigates the effects of integrating a soft exoskeleton with a myoelectric prosthesis on upper-limb muscle fatigue and user experience. Nine non-disabled participants performed four functional tasks: drinking from a cup, using a Fork, lifting a box, and reaching overhead, using the prosthesis alone and in combination with the exoskeleton. Muscle activity was recorded via surface electromyography, and perceived exertion was measured using the Borg scale. Kinematics and workload were also assessed through motion capture and the NASA-TLX questionnaire. Usability was evaluated using the System Usability Scale (SUS). Results showed that exoskeleton assistance significantly reduced muscle activation, particularly in the Deltoid, Biceps, and Triceps Lateral Head during the Lift task, with RMS reductions up to 64 % and large effect sizes. Perceived exertion slopes decreased across all tasks, with some instances showing stabilization or reduction during activity. Kinematic analysis indicated minimal impact on shoulder range of motion, with slight adjustments in internal/external rotation remaining within physiological norms. NASA-TLX scores suggested reduced physical demand and effort, and SUS responses indicated moderate usability with room for improvement. These findings demonstrate that soft exoskeletons can effectively unload muscles and reduce fatigue during prosthesis use, highlighting their potential to enhance endurance, task performance, and user comfort. Future work should extend assistance to additional joints and evaluate the system with upper-limb amputees in real-world scenarios.**

***Index Terms*— Exoskeleton, fatigue analysis, prosthesis, weight compensation.**

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## I. Introduction

IN 2024, ACCORDING to DATASUS, approximately 33,000 upper limb amputations were performed in Brazil [1]. In the United Kingdom (UK), the Institute for Health Metrics and Evaluation (IHME) reported that in 2021, the prevalence of upper-limb amputations was close to 54,000 [2]. These statistics underscore the pressing need for effective prosthetic solutions to improve the quality of life for amputees. Prosthetic devices are primarily designed to alleviate the psychological impact of limb loss and assist in performing activities of daily living (ADLs) [3].

Prostheses are generally categorized into three main types: passive, body powered, and electrical powered. Passive prostheses are primarily used as cosmetic replacements. Body-powered prostheses use a system of harnesses and cables to capture user movements and control the device. On the other hand, electrically powered prostheses rely on electronic components for actuation, typically, motors. These devices often use surface electromyography (sEMG) signals to control movements [4].

Despite significant technological progress, weight remains one of the most commonly reported issues for all types of prosthetic devices [5]. Because users typically wear prostheses for 6–8 hours per day, prolonged use can lead to muscle fatigue. This fatigue affects the quality of sEMG signals, which are used for control, and affects the performance of the prosthesis [6]. These challenges can lead to decreased usage and even abandonment of the device, with approximately 20 % of amputees discontinuing prosthesis use [7].

Some current techniques for mitigating the impact of fatigue on sEMG signals involve the use of machine learning (ML) algorithms or integrating sEMG signals with other physiological signals. An approach using ML involves classifying unintended movements [8]. However, if a new or untrained movement occurs, the algorithm can produce errors. Another strategy involves combining sEMG signals with electroencephalography (EEG) data to compensate for fatigue-induced control issues. Although promising, EEG-based systems still face several limitations, including signal acquisition challenges, low user adaptability, high computational demands, and limited portability [9]. Most state-of-the-art studies focus on signal processing. Few studies have proposed physical devices to prevent fatigue. One example is found in [10], where vibration-based feedback sensors were used to help users adjust their muscle contraction. The results showed that users exerted less effort when gripping with feedback, which contributed to the delay in the onset of fatigue.

Robotic exoskeletons have been highlighted in literature as promising alternatives for preventing muscle fatigue [11]. These devices are designed to be worn on the human body to enhance the user's physical capabilities during ADLs [12] or working activities [13]. For example, a passive upper-limb exoskeleton was developed and experimentally validated to reduce trunk and shoulder muscle activation during static and quasi-dynamic tasks [13]. Similarly, a wearable passive lower-limb support exoskeleton designed for industrial workers demonstrated significant reductions in quadriceps activation and perceived exertion during prolonged lifting and standing tasks [14]. Moreover, a lightweight exoskeleton chair, mechanically designed and experimentally verified, has been shown to assist sit-to-stand movements and lessen lower-limb loading in healthy subjects [15]. These studies underscore the potential of passive support devices across different body regions. In this context, an emerging area of research involves the integration of an exoskeleton to assist movement during prosthesis use. Most studies in this area have focused on lower limbs, such as the integration of a hip exoskeleton with a lower-limb prosthesis [16], [17]. The results from these studies showed a reduction in the metabolic cost of walking, thus improving user mobility.

Regarding upper limbs, as far as the authors' knowledge, only one study has explored this integration. In [18], a system combining an upper-limb exoskeleton and prosthesis was introduced, where the exoskeleton works as the prosthetic socket. The results demonstrated that the device could compensate for 50 % of the required force in dynamic tasks and 100 % in static scenarios. Some areas for improvement include reducing the overall weight of the device and enhancing the actuation system. Notably, the system presented in [18] uses rigid robotics, leaving the application of soft-robotics largely unexplored in this field.

Considering the negative impact of prosthetic weight on device usage time, muscle fatigue, and control, this study proposes the use of an exoskeleton to compensate for the weight of the prosthesis and to validate whether it significantly reduces muscle fatigue during prolonged use. Muscular fatigue is known to begin manifesting after approximately five minutes of prosthesis use [19]. For the test, the latest version of the PrHand prosthesis was used [20]. The device weighs 550 g, which is lighter than the human forearm, typically ranging from 755 g to 1,400 g [21].

## II. Devices

### A. Prosthesis

This study used a third-generation PrHand prosthesis (see Fig. 1), as detailed in [20]. The device integrates soft-robotics and compliant mechanisms to enable natural finger movement. Its elastic joints allow finger extension via an internal flexible tendon, while finger flexion is driven by inelastic tendons actuated by a Dynamixel motor MX-106 (Robotis, Republic of Korea). These tendons run from each fingertip and converge into a single tendon connected to the motor. The compliant finger design enables multiple degrees of freedom (DOFs), including flexion, extension, abduction, and adduction. To enhance grip performance, the fingers were coated with silicone to increase friction during object manipulation.

The prosthesis is controlled via the Robot Operating System (ROS), running on a Raspberry Pi 3 single-board computer. A sEMG sensor placed over the extensor digitorum muscle was used to command the opening and closing of the PrHand. For the experiment, a custom bypass socket was developed to allow non-amputee participants to perform the activities using the prosthesis [22]. The socket was designed to avoid restricting natural elbow movement and to accommodate individual user dimensions, enabling proper alignment and functionality during the tasks.

### B. Exoskeleton

For the test, an upper limb exosuit was employed, a wearable device designed to assist elbow joint movement during flexion and extension, primarily by reducing the load on the biceps muscle (see Fig. 1) [23]. The exosuit delivers active, cable-driven torque assistance using two Dynamixel MX-106T DC motors (one per arm) mounted on a modular frame around the lumbar region. Each motor reels in a steel cable anchored to a 3D-printed, handcuff-shaped forearm attachment, generating up to 5.6 Nm of torque and thereby sharing the load that would otherwise be borne by the biceps. Weight compensation is achieved through a continuous load path: tension from the forearm cuff is routed over breathable, backpack-style shoulder straps into the rigid exospine frame, which conforms to the natural lumbar curvature and consists of four articulated dorsal segments connected by a steel cable. This structure transmits forces into the lumbar belt and down into the pelvis and lower back, effectively bypassing the elbow joint, while still allowing up to 8° of abduction/adduction per segment and up to 51° of sagittal flexion/extension.

The exosuit interfaces with the body at three main contact points for seamless interaction during movement. An adjustable padded lumbar belt secures the exospine frame around the waist, matching spinal curves and supporting natural torso bending. Shoulder straps stabilize cable routing without restricting thoracic motion, and padded forearm cuffs grip the mid-ulnar shaft comfortably while transmitting assistive torque. At the low level, the Dynamixel torque mode was activated to monitor motor current and estimate the

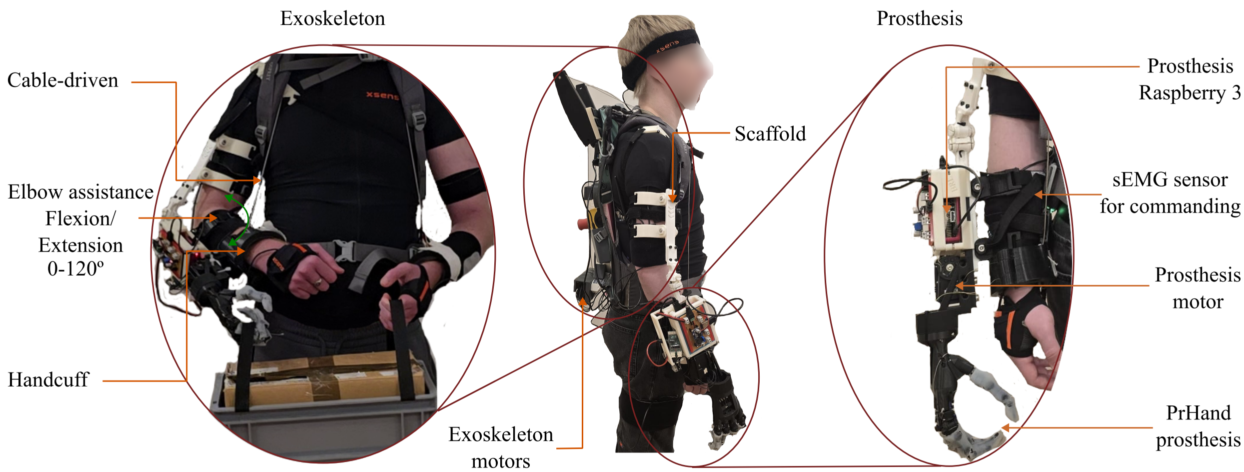


Fig. 1. A participant wearing the PrHand prosthesis and the assistive exoskeleton. The prosthesis is mounted on the right arm, while the exoskeleton's mechanical structure and control system are supported by a backpack.

generated torque. At the mid level, an impedance controller used the desired angular position and velocity, estimated with an Inertial Measurement Unit (IMU), to compute the torque, which was then applied by the actuation system to the user's forearm. However, for the present study, this control strategy was simplified: the low-level torque controller remained active, while flexion and extension phases were manually triggered by an operator. Under this configuration, the exosuit still provided active torque assistance, reducing the effective weight experienced at the elbow by transmitting cable tension through the exospine into the pelvis. This ensured that the user experienced support against gravity while natural elbow and torso motion remained unrestricted. The exosuit operates within a range of motion from 0° to 120°, and the total weight of the exosuit used in the load-bearing tests was 2.5 kg. The processing and control system is based on a Raspberry Pi 4 running ROS Melodic.

## III. Methodology

### A. Participants

This study involved the voluntary participation of nine subjects (four female and five male) [24]. Inclusion criteria required participants to be healthy individuals aged 18 to 50 years and in a non-fatigued state, as determined by the Multidimensional Fatigue Inventory (MFI) [25], which assesses five dimensions of fatigue: general fatigue, physical fatigue, mental fatigue, decreased motivation, and reduced activity. Participants with a history of arm fractures, musculoskeletal or systemic disorders, or any known impairment of postural control or motor function were excluded. All participants were informed of the objectives and procedures of the study and provided written informed consent prior to participation. This research has been approved by CATE Research ethics committee (Reference number 13312133).

### B. Instrumentation

*1) Surface Electromyography (sEMG):* Five sEMG electrodes (Delsys Incorporated, USA) were used to measure muscle activity in the Trapezius (T), Deltoid (D), Biceps (B), Triceps Short Head (TSH), and Triceps Lateral Head (TLH). These muscles were selected due to their involvement in upper-limb motion and compensatory strategies typically adopted by prosthesis users [26]. Electrode placement followed SENIAM guidelines [27] to ensure reproducibility and minimize signal cross-talk. Prior to the task trials, a Maximum Voluntary Contraction (MVC) protocol was conducted. Each participant performed three isometric contractions lasting three seconds, with 30-second rest intervals in between. The procedure was repeated three times per muscle to ensure effort consistency and improve normalization reliability. A sampling frequency of 2000 Hz was used for sEMG data acquisition to capture high-resolution signals.

*2) Kinematics:* In addition, the XSens Awinda suit (Movella, Netherlands) was used to capture full-body kinematics during the tests. This MVN Biomechanical motion capture system consists of 10 IMUs positioned on the head, shoulders, upper arms, forearms, hands, sternum and pelvis for building the upper body's biomechanical model. The sensors were wirelessly attached to the body using a shirt and adjustable straps, and their placement was verified before each trial.

### C. Experimental Protocol

The experiment was conducted in two separate sessions on different days. In the first session, participants performed the tasks using only the prosthesis. In the second session, both the prosthesis and the cable-driven exoskeleton were used. This design allowed for a comparative evaluation of muscle fatigue, kinematics, and user experience under both conditions.

After the instrumentation, during both sessions, participants were fitted with the prosthesis using a four-point strapping system: two straps on the forearm, two on the upper arm, and an additional chest strap to stabilize and distribute the device weight. Before the tasks, a calibration procedure required participants to perform three brief fist closures, each lasting one second and separated by ten seconds of rest, to determine muscle activation thresholds for prosthetic hand control. In the second session, after fitting the prosthesis, the exoskeleton was

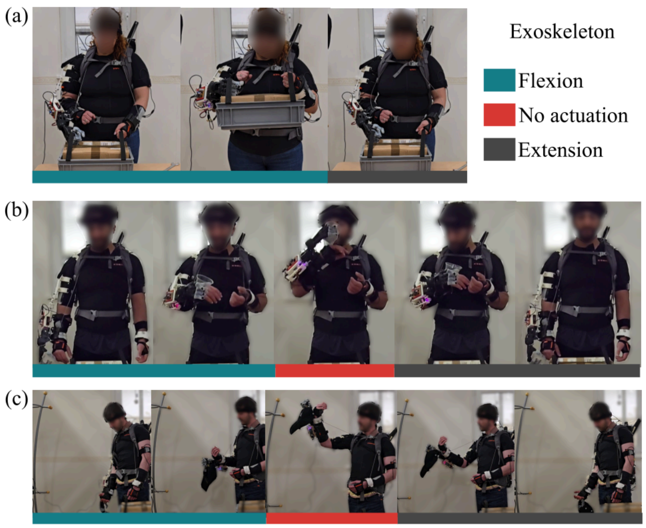


Fig. 2. Sequence of assisted and unassisted phases during each functional task with the exoskeleton activated. Cyan indicates the interval in which the exoskeleton provides assistance (0°–120° of elbow flexion). Pink marks the phase where the participant completes the movement using their own effort, and gray represents the passive extension phase. (a) Lift activity. (b) Cup activity (same sequence for Fork). (c) Reach overhead activity.

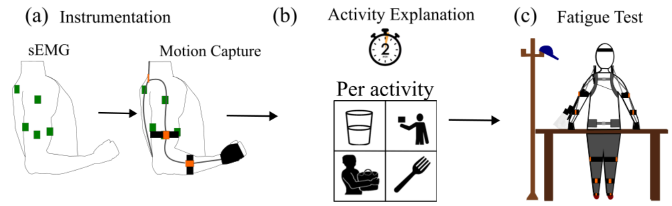


Fig. 3. Experimental setup for assessing muscle fatigue reduction using the exoskeleton during daily activities. (a) Instrumentation. (b) Explanation of the functional activities. (c) Fatigue assessment.

also installed. Since participants had varying arm lengths, the exoskeleton cable-driven system and motors were individually calibrated to assist elbow flexion and extension within a range of 0° to 120° (Fig. 1).

The experimental tasks consisted of four functional movements adapted from the Activities Measure for Upper Limb Amputation (AM-ULA) [28]: bringing a metal fork to the mouth to simulate eating, lifting a 200 ml empty plastic cup to the lips, picking up a box weighing 1–2 kg, and reaching overhead to retrieve a cap from a 2-meter-tall hanger. Based on the approach of Bangaru et al. [29], who evaluated muscle fatigue across a circuit of industrial activities, and considering evidence that muscle fatigue typically emerges after approximately five minutes of continuous prosthesis use [19], the present protocol was designed to include four activities of daily living, each performed for two minutes. This structure resulted in a total of eight minutes of continuous use, exceeding the duration after which muscle fatigue typically develops, while allowing the assessment of the ExoProsthesis during different functional activities.

Before starting the test, the operator explained each activity to the participants to ensure consistent task execution. Given that the exoskeleton provided assistance from 0° to 120° of elbow flexion, participants had to complete some portions of the movement without assistance. Specifically, for the box task, which required lifting to approximately 120°, the exoskeleton fully supported the movement (see Fig. 2 part a). In contrast, for the Cup and Fork tasks, the exoskeleton lifted the arm up to 120° of elbow flexion, and participants were responsible for completing the fine motor action of bringing the items to the lips (see Fig. 2 part b). During the Reach overhead task, the exoskeleton again assisted up to 120°, while participants raised their arms further to reach the object on the tall hanger (see Fig. 2 part c).

During the sessions, participants rated their fatigue every 30 seconds using the Borg Rating of Perceived Exertion (RPE) scale, ranging from 0 (no exertion) to 10 (maximal exertion) [30]. In the prosthesis-only session (*PO*), participants were allowed two brief rest pauses; a third pause due to high fatigue resulted in termination of the test. In the ExoProsthesis session (*EP*), the exoskeleton was manually activated when participants reported a fatigue level of 6, corresponding to "high" to "very high" exertion [23], and assistance continued until the fatigue level decreased to 3. This activation/deactivation cycle was repeated across all four tasks. Level 6 was chosen because participants typically begin to experience substantial reductions in muscular strength and compensatory movements at this point, which can reduce efficiency and increase injury risk [30], [31]. The operator manually controlled the assistance for both flexion and extension. Fig. 3 illustrates the experimental setup and the procedure for exoskeleton activation during tasks.

Following each session, participants completed the NASA Task Load Index (NASA-TLX) [32] to assess perceived workload during the tasks. This questionnaire was applied in both sessions, *PO* and *EP*, to evaluate cognitive and physical demand, effort, and performance perception under both conditions. In addition, only after the second session *EP*, participants completed the System Usability Scale (SUS) [33], a standardized questionnaire designed to measure perceived usability of the combined assistive system.

## D. Data Processing

*1) Video Analysis:* All sessions were recorded using two digital video cameras placed in front of the user and the other laterally, to capture the movements of the upper extremities of the participant and the behavior of the device throughout the tasks. These recordings were reviewed to extract qualitative and quantitative information on device performance, participant behavior, and to annotate task performance in terms of whether each task was effectively completed.

Prosthesis control failures were also analyzed, defined as involuntary openings or closings of the hand occurring when the participant reported a Borg fatigue score greater than 5. Each malfunction was recorded and classified by activity and condition (*PO* and *EP*). Participant pauses were taken from video recordings. A pause was defined as any voluntary interruption of the task initiated by the participant. These annotations complemented the subjective fatigue data and allowed identification of complete task discontinuation due to excessive fatigue, particularly in the *PO* condition.

*2) Borg Scale:* The Borg values recorded during the test were used to evaluate the variation in perceived exertion while performing activities. The variation in perceived exertion was calculated for each activity per participant, under two conditions: using the prosthesis alone, and using the prosthesis in combination with the exoskeleton. For each participant and each condition, the initial RPE value was subtracted from the subsequent values to calculate the change in perceived exertion over time. A linear regression was applied to these values to obtain the slope, representing the rate of increase in perceived effort throughout the task. This process was repeated for each of the four activities. Finally, the mean slope was computed across participants for each condition, enabling a comparison of exertion dynamics between the prosthesis-only and exoskeleton-assisted tasks.

*3) XSens Processing:* The kinematic data recorded with the XSens Awinda suit were processed using the MVN Analyze software and exported for analysis. From the shoulder joint angles, three DOFs were selected: Flexion/Extension (F/E), Abduction/Adduction(A/A), and Internal/External rotation (I/E).

For each repetition of the functional tasks, the minimum, maximum, range (max-min), and RMS values were calculated for each degree of freedom. These metrics allowed for the quantification of movement, amplitude, and variability during task execution.

The primary objective of this analysis was to evaluate whether the cable routing of the exoskeleton constrained the natural scapulohumeral rhythm. By comparing shoulder kinematics across the *PO* and *EP* sessions, potential restrictions or alterations in shoulder mobility could be identified.

*4) NASA Task Load:* The NASA-TLX evaluates subjective workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration. Participants rated each dimension on a scale from 0 to 20 in increments of 1. In this study, the raw NASA-TLX method was used, where scores for each dimension were averaged to compute a composite workload score. As the performance dimension is inversely rated, it was reversed during processing to align it with the direction of the other dimensions. This correction ensures that higher values consistently reflect a greater workload. The final composite score reflects the overall perceived workload of the participant during the session using the prosthesis and exoskeleton system [32].

*5) System Usability Scale:* The SUS consists of ten statements rated on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). To process the SUS scores, responses to odd-numbered items were adjusted by subtracting 1 from the score, while responses to even-numbered items were adjusted by subtracting the score from 5. The adjusted values for all items were then summed and multiplied by 2.5 to obtain a total usability score ranging from 0 to 100. Higher scores indicate better perceived usability. The resulting scores were interpreted using standard SUS benchmarks [33].

### *E. Statistic Analysis*

The null hypothesis stated that there were no significant differences between conditions when using only the prosthesis compared to using the prosthesis in combination with the exoskeleton. Data normality was assessed using the Shapiro–Wilk test. Depending on the results, either a Paired Samples t-test (for normally distributed data) or a Wilcoxon Signed-Rank test (for non-normal data) was applied. All statistical analyses were performed in R Studio, with a significance level set at 5 % ($p < 0.05$).

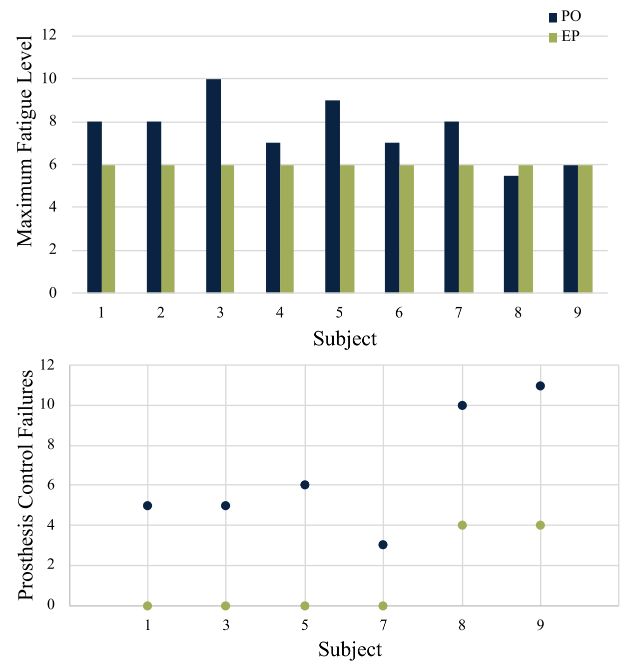


Fig. 4. Video analysis results of *PO* and *EP* sessions. (a) Maximum fatigue level reached during each session per participant. (b) Number of prosthesis control failures.

In addition, shoulder kinematics obtained from the XSens system were analyzed across three conditions: PO (Prosthesis Only), *EP_P* (ExoProsthesis with only the Prosthesis active), and *EP_EP* (ExoProsthesis with Exoskeleton and Prosthesis active). Normality was assessed for each variable per DOF. When data followed a normal distribution, a repeated-measures ANOVA was applied; otherwise, the non-parametric Friedman test was used. Significant effects were further examined with post-hoc pairwise comparisons (paired t-tests with Bonferroni correction or Wilcoxon Signed-Rank tests with Holm correction). The tests were performed for maximum, minimum, range, and RMS values per DOF.

This analysis enabled us to evaluate whether the presence and activation of the exoskeleton significantly influenced shoulder kinematics and, in particular, whether cable routing restricted the natural scapulohumeral rhythm.

## IV. Results

Video analysis revealed notable differences between *PO* and *EP* conditions in fatigue progression, prosthesis reliability, and task interruptions. As shown in Fig. 4a, maximum fatigue levels were generally higher in the *PO* condition: seven out of nine participants reported greater fatigue when using only the prosthesis, one reported equal levels, and only one experienced slightly higher fatigue with the exoskeleton. These findings

TABLE I

ACTIVITIES IN WHICH THE EXOSKELETON WAS ACTIVATED FOR EACH PARTICIPANT. ACTIVATION OCCURRED WHEN A PARTICIPANT REPORTED A FATIGUE LEVEL OF 6 ON THE BORG SCALE

| Individual | Activities | | | |
|---|---|---|---|---|
| | Cup | Fork | Lift | Reach |
| 1 | ■ | ■ | ■ | |
| 2 | ■ | ■ | ■ | ■ |
| 3 | | | ■ | |
| 4 | | ■ | | |
| 5 | | | ■ | ■ |
| 6 | | | ■ | |
| 7 | ■ | ■ | ■ | |
| 8 | ■ | | ■ | ■ |
| 9 | | | | ■ |

TABLE II

PERCENTAGE CHANGE (PC %) IN RMS VALUES FOR *EP_EP* RELATIVE TO *EP_P* ACROSS FOUR FUNCTIONAL TASK. NEGATIVE VALUES INDICATE A REDUCTION IN MUSCLE ACTIVATION WITH THE EXOSKELETON. STATISTICALLY SIGNIFICANT DIFFERENCES (P <0.05) ARE SHOWN IN BOLD

| | Cup | | Fork | | Lift | | Reach | |
|---|---|---|---|---|---|---|---|---|
| | PC (%) RMS | p | PC (%) RMS | p | PC (%) RMS | p | PC (%) RMS | p |
| T | -61.53 | 0.20 | -57.17 | 0.03 | -35.53 | 0.07 | -10.73 | 0.87 |
| D | -53.94 | 0.12 | -38.90 | 0.25 | -64.16 | 0.02 | -27.71 | 0.14 |
| B | -45.02 | 0.07 | -56.48 | 0.25 | -50.95 | 0.07 | -18.33 | 0.63 |
| TSH | -41.28 | 0.15 | -21.49 | 0.79 | -40.63 | 0.02 | -19.22 | 0.10 |
| TLH | -46.65 | 0.14 | -40.71 | 0.21 | -48.41 | 0.02 | -31.07 | 0.09 |

suggest that exoskeleton assistance effectively reduced fatigue during task execution.

Fig. 4b shows prosthesis control failures, which occurred predominantly in the *PO* condition, with Subjects 5, 8, and 9 exhibiting up to 11 failures. During *EP*, failure rates were reduced or eliminated in nearly all cases. Participants 2, 4, and 6 exhibited no control failures across any session. Additionally, two participants required voluntary pauses during *PO* due to high fatigue, whereas all completed *EP* sessions without interruption, supporting the fatigue-mitigating effect of the exoskeleton. Overall, task performance was only affected by prosthesis control failures; in all other cases participants were able to complete the tasks without difficulty.

Since exoskeleton activation was triggered at Borg level 6 and task order was randomized, the device was not engaged in the same activity for all participants. Table I summarizes the tasks that prompted exoskeleton assistance, with Lift being the most frequently assisted activity.

Fig. 5 presents the activation levels of the RMS muscles for the five muscles in the upper limb T, D, B, TSH, and TLH, during the performance of four functional tasks: Cup, Fork, Lift, and Reach. Comparisons were made between two conditions during the second session: *EP_P* and *EP_EP*.

Across most muscles and tasks, RMS values were consistently lower in the *EP_EP* condition compared to the *EP_P* condition, indicating a reduction in muscular effort. The Percentage of Change (PC), calculated relative to the *EP_P* condition, highlights this trend (see Table II), with the most substantial decreases observed in the Lift and Cup activities. For example, during the Lift task, the D showed a 64.16 % reduction in RMS, and the TLH showed a 40.63 % reduction; both results were statistically significant (p = 0.02). Similarly,

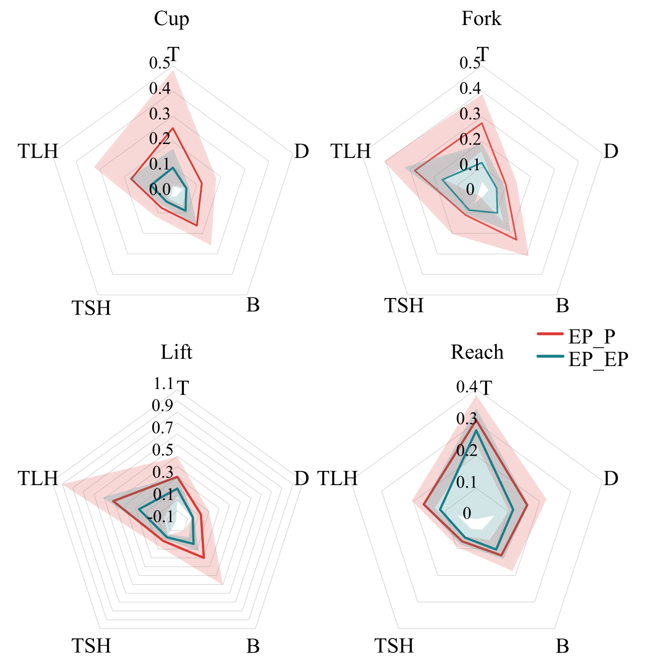


Fig. 5. RMS result for each activity across five muscles: Trapezius (T), Deltoid (D), Biceps (B), Triceps Short Head (TSH), and Triceps Lateral Head (TLH). Comparisons are shown in the ExoProsthesis session between *EP_P* and *EP_EP* conditions. Shaded areas are the standard deviation.

TABLE III

MEAN SLOPE OF CHANGE IN PERCEIVED EXERTION (ΔBORG/ΔT) ACROSS FOUR DAILY LIVING ACTIVITIES UNDER TWO CONDITIONS: *EP_P* AND *EP_EP*. VALUES REPRESENT MEAN ± STANDARD DEVIATION. THE PERCENTAGE CHANGE (PC %) IS CALCULATED RELATIVE TO THE *EP_P* CONDITION

| | | ΔBorg/Δt | PC(%) |
|---|---|---|---|
| Cup | *EP_P* | 0.02 ± 0.01 | -348.80 |
| | *EP_EP* | -0.05 ± 0.03 | |
| Fork | *EP_P* | 0.02 ± 0.01 | -142.45 |
| | *EP_EP* | -0.01 ± 0.00 | |
| Lift | *EP_P* | 0.04 ± 0.02 | -202.99 |
| | *EP_EP* | -0.04 ± 0.02 | |
| Reach | *EP_P* | 0.03 ± 0.02 | -187.80 |
| | *EP_EP* | -0.03 ± 0.01 | |

during the Cup activity, the T, D, and B exhibited reductions of 61.53 %, 53.94 %, and 45.02 %, respectively. Although these differences were not statistically significant, all had Cohen's d values above 0.8, indicating large effect sizes.

In the Fork task, the T muscle showed a statistically significant RMS reduction of 57.17 % (p = 0.03), with a large effect size (Cohen's d > 0.82), indicating a substantial decrease in muscle activation when the exoskeleton was used. Although the B muscle exhibited a similar RMS reduction of 56.48 %, this difference was not statistically significant (p = 0.25), despite also presenting a large effect size (Cohen's d > 0.82). In the Reach task, consistent RMS reductions were observed across muscles (e.g., 31.07 % in the TLH), but none of these reached statistical significance. This was the activity with lower PC.

Table III presents the average rate of change in perceived exertion (ΔBorg/Δt) for each activity under two experimental conditions *EP_P* and *EP_PE*. Across all four activities, a

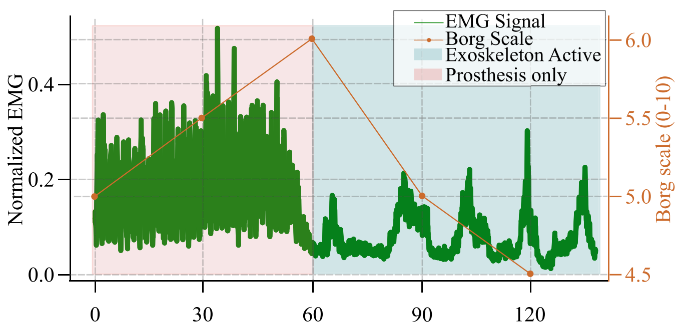


Fig. 6. Example of a normalized sEMG signal from the biceps (green line) recorded during a functional task. Muted Pink-Red shaded areas represent periods when only the prosthesis was used, while Deep Teal-Blue shaded areas indicate when the exoskeleton was active. The orange line corresponds to the participant's reported values on the Borg Rating of Perceived Exertion (RPE) scale, recorded every 30 seconds.

consistent reduction in the slope of perceived exertion was observed when the exoskeleton was used.

Specifically, during the Cup task, the slope decreased from 0.02 $\pm$ 0.01 in the *EP_P* condition to –0.05 $\pm$ 0.03 in the *EP_EP* condition, corresponding to a 348.80 % reduction. Similar trends were observed for the Fork task (–142.45 %), Lift (–202.99 %), and Reach (–187.80 %). Notably, in the *EP_EP* condition, the slope values were either negative, suggesting a stabilization or reduction in perceived exertion over time, indicating effective fatigue mitigation by the exoskeleton.

Fig. 6 shows an example of the sEMG signal for a participant, showing a higher muscle activation over time during task execution with the prosthesis alone. Upon reaching a Borg scale rating of 6, the exoskeleton was activated, resulting in a noticeable reduction in both perceived exertion and sEMG amplitude. This trend suggests that the exoskeleton effectively reduces muscular load when user fatigue increases.

In addition to muscle activity and RPE, shoulder kinematics were evaluated under three experimental conditions: *PO*, *EP_P*, and *EP_EP*. Fig. 7 presents the maximum, minimum, range, and RMS values for F/E, A/A, and I/E.

Overall, the results showed that the integration of the exoskeleton *EP* did not significantly alter the range of motion compared to *PO*, with similar kinematic profiles across the three directions. Nevertheless, significant differences were observed in specific parameters, particularly in I/E Rotation, suggesting slight postural adjustments related to the exoskeleton assistance. RMS analysis confirmed that movement variability remained within a physiologically normal range, supporting that exoskeleton engagement did not compromise shoulder functional mobility.

To assess perceived workload, participants completed the NASA-TLX questionnaire after each session. Although there were no statistically significant differences between the evaluated variables across the two conditions, some trends can be observed in Fig. 8. Physical demand and effort scores were lower in the *EP* condition compared to *PO*, suggesting a reduction in physical strain when using the exoskeleton. Mental demand, temporal demand, and performance were similar across conditions. Notably, frustration levels were slightly higher with the ExoProsthesis. The mean overall workload

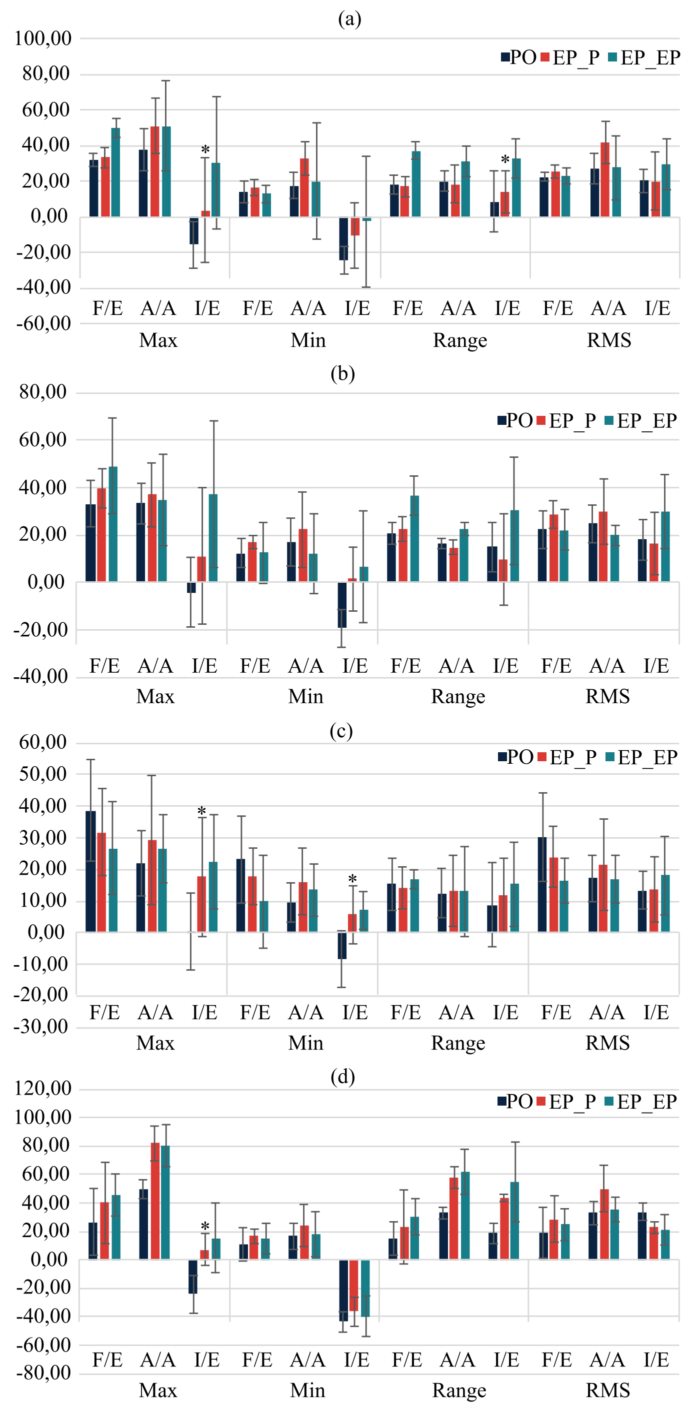


Fig. 7. Shoulder kinematics during functional tasks under three conditions: *PO*, *EP_P*, and *EP_EP*. Maximum, minimum, range, and RMS values are presented for flexion/extension (F/E), abduction/adduction (A/A), and internal/external rotation (I/E). Significant differences were observed in I/E parameters (*$p < 0.05$).

score was 9.74 $\pm$ 1.99 for the *PO* session and 9.07 $\pm$ 3.00 for the *EP* session.

Table IV summarizes participants' responses to the usability questionnaire adapted from the SUS, focusing on their interaction with the prosthesis exoskeleton system. Overall, the results indicate a generally positive perception of the system's usability. However, participants did not express a strong willingness to use the system frequently (2.00 $\pm$ 1.25)

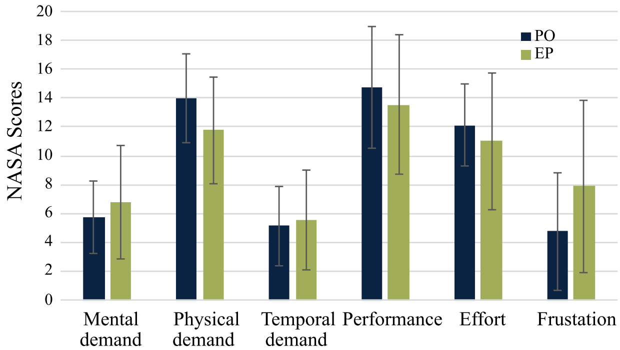


Fig. 8. Mean scores and standard deviations for the six NASA-TLX workload dimensions reported after task execution with the prosthesis-exoskeleton system. Higher values indicate greater perceived demand or lower performance.

TABLE IV
MEAN AND STANDARD DEVIATION SCORES FOR USER RESPONSES TO THE USABILITY QUESTIONNAIRE BASED ON THE SUS

| Questions | Mean Score |
|---|---|
| I would like to use the exoskeleton with the prosthesis often | 2.00 ± 1.25 |
| I think the exoskeleton with the prosthesis is complex to use | 2.33 ± 1.05 |
| I think the exoskeleton with the prosthesis is easy to use | 1.89 ± 0.87 |
| I think I would need technical support to use this system effectively | 2.44 ± 1.17 |
| I think the functionalities of the systems are well integrated | 2.33 ± 0.47 |
| I think the functionalities of the systems are not consistent | 3.11 ± 0.57 |
| I think most users can quickly learn to use the exoskeleton with the prosthesis | 3.00 ± 0.94 |
| I think most users have difficulties learning to use the exoskeleton with the prosthesis | 2.78 ± 0.92 |
| I am confident when using both devices | 1.56 ± 1.07 |
| I need to learn more background information of the exoskeleton with the prosthesis before use | 2.67 ± 0.94 |

and reported limited confidence while using it (1.56 ± 1.07). Users also perceived the system as not particularly complex (2.33 ± 1.05).

Moderate scores were observed for items related to the need for technical support (2.44 ± 1.17) and prior background knowledge (2.67 ± 0.94), suggesting that some initial guidance might enhance the onboarding process. Notably, participants agreed that most users could learn to use the system quickly (3.00 ± 0.94). These findings support the system's perceived usability and suggest that only minimal additional support or training may be required for new users. The overall usability score was 60.28 ± 14.02, indicating that users found the exoskeleton useful but that improvements are still needed.

## V. DISCUSSION

This study aimed to evaluate whether integrating a soft exoskeleton with a myoelectric prosthesis could effectively reduce muscle fatigue during prolonged use. The results provide compelling evidence that the exoskeleton contributes to lower muscle activation and reduced perceived exertion across several functional tasks, consistent with prior findings that wearable robotic assistance can reduce muscular effort during physical activities [34].

The results presented in Fig. 4 provide initial evidence of the fatigue-reducing effect of the soft exoskeleton when used in combination with a myoelectric prosthesis. Specifically, the maximum fatigue level reported during each test was generally higher in the *PO* session compared to the *EP* session. Prosthesis control failures associated with fatigue were considerably reduced, indicating improved prosthetic performance. Notably, two participants required voluntary pauses to complete the prosthesis-only session due to high fatigue levels, whereas all participants completed the ExoProsthesis session without interruption. These findings suggest that the mechanical assistance provided by the exoskeleton contributed to reducing fatigue accumulation and performance degradation, thereby enhancing endurance during task execution.

The exoskeleton was most frequently activated during the Lift task, indicating that this activity imposed the greatest physical demand. Correspondingly, the largest reductions in RMS muscle activity were observed during Lift, particularly in the D, B, and TLH muscles. These reductions were statistically significant and accompanied by large effect sizes, suggesting a meaningful physiological benefit from exoskeleton support [35]. Lift was also the task with the highest level of exoskeleton assistance, reinforcing the link between mechanical support and muscular unloading.

A general reduction in muscle activity was observed across all tasks when the exoskeleton was active. Although statistically significant differences were only evident in the Lift task, this may be due to the limited number of valid samples in other tasks. Specifically, more participants reached the activation threshold for exoskeleton engagement during Lift, increasing statistical power for this condition. This aligns with the physical demands of the task, which involved lifting the heaviest object in the protocol, further compounded by the combined weight of the prosthesis and exoskeleton.

Despite the limited number of effective samples in the other tasks ($n = 4$ per condition), Cohen's d consistently exceeded 0.8, indicating large effect sizes across the board. This suggests that the absence of statistical significance in some activities is more likely due to sample size limitations rather than the absence of a true effect [36]. Notably, the T and TLH muscles, commonly engaged in compensatory movements due to prosthesis weight, showed consistent reductions in activation. This aligns with research indicating that minimizing compensatory activity may reduce the risk of secondary injuries in prosthesis users [5], [37].

These findings provide further biomechanical insight into how the exoskeleton redistributes muscle effort across joints and muscle groups. Significant reductions in RMS values during the Lift task, particularly in D, B, and TLH, demonstrate effective unloading of muscles primarily involved in elbow flexion and shoulder stabilization. The simultaneous reductions in biceps and triceps activity suggest that assistance does not merely shift demand from one muscle group to another, but allows an overall reduction in co-contraction and global muscle strain. This pattern aligns with improved mechanical efficiency during lifting, a task known to impose high biomechanical demands on the upper extremity [38].

The Fork task also showed significant reductions in T activity, indicating that even during precision-oriented, moderate-load tasks, the exoskeleton contributes to unloading stabilizing muscles involved in elbow extension and positional control. This may point to its role in reducing gross motor effort while supporting fine motor stability.

Consistent decreases in T and TSH muscle activation across tasks suggest a reduction in compensatory shoulder elevation strategies, commonly observed in prosthesis users to counteract distal weight and limited control [37]. Reducing such compensatory demand is biomechanically significant, as it may help prevent secondary musculoskeletal problems [39].

The Reach task showed the smallest percentage reduction in RMS values, likely due to mechanical constraints of the exoskeleton, which primarily assists elbow flexion up to approximately 120°. As the Reach task required overhead movement, participants had to rely more on shoulder and upper arm muscles beyond the exoskeleton's assistive range. This highlights an area for future design improvement, specifically incorporating shoulder assistance to support a broader range of upper-limb movements relevant to daily living [40].

Taken together, these results suggest that the soft exoskeleton promotes biomechanically more efficient task execution by reducing the need for both primary muscle activation and compensatory postural adjustments. This may translate into long-term benefits such as improved endurance, reduced risk of overuse injuries, and enhanced motor performance, particularly during high-load or repetitive activities.

Shoulder kinematics were analyzed to assess whether exoskeleton use altered natural joint mobility. F/E and A/A ranges remained comparable across conditions, suggesting that the load-transfer mechanism did not impede primary shoulder movements. Significant differences were observed in I/E rotation, likely reflecting compensatory adjustments induced by the exosuit cable routing and frame support. Prior studies have shown that upper-limb exoskeletons can promote more externally rotated shoulder postures, requiring compensatory internal forearm rotation to maintain hand orientation [41]. Importantly, observed I/E differences remained within physiologically normal limits, indicating that the exosuit influenced posture without compromising functional mobility.

Subjective data further supported physiological findings. The slope of the Borg scale decreased markedly with exoskeleton assistance and, in some cases, became negative, indicating stabilization or reduction in perceived exertion over time. For example, during the Cup task, perceived exertion declined despite ongoing activity, possibly due to reduced postural effort in the shoulder. Table III shows a decreasing trend in perceived exertion during the short activity period, likely reflecting the immediate effect of mechanical assistance rather than cumulative fatigue. Similar trends have been reported in other upper-limb assistive devices, where users experience an initial decrease in perceived exertion due to effective load compensation [42], [43]. Future work will involve longer-duration studies to understand how RPE evolves under sustained exosuit use.

An illustrative sEMG trace demonstrated an immediate reduction in muscle activity upon exoskeleton engagement, which coincided with a decrease in perceived exertion. A progressive increase in exertion was observed until reaching the activation threshold (level 6 on the Borg scale), followed by a reduction once assistance was provided. Responsiveness to movement and user intent is essential for maintaining comfort during dynamic tasks and enhancing user acceptance [44].

Although NASA-TLX scores did not show statistically significant differences, trends indicate lower physical demand and effort ratings in the *EP* condition, consistent with observed reductions in muscle activation and RPE. Mental and temporal demands, as well as perceived performance, were similar across conditions, suggesting that the exoskeleton does not introduce additional cognitive load. Frustration levels were slightly higher with *EP*, likely reflecting the novelty of the device or the learning curve for untrained users. Overall, the slightly lower global workload score in *EP* ($9.07 \pm 3.00$) versus *PO* ($9.74 \pm 1.99$) supports the potential of the exoskeleton to improve comfort during task performance with continued user familiarization.

Complementary insights were gained from the SUS, suggesting moderate usability. Participants found the system useful but reported low confidence and limited inclination for frequent use. The system was generally not overly complex, and participants agreed that most users could learn to operate it quickly. These results align with prior usability studies of upper-limb assistive technologies, highlighting the importance of early training and intuitive control schemes [33].

Together, these findings support the feasibility of using soft exoskeletons to actively reduce muscle fatigue in upper-limb prosthesis users. Unlike approaches that target fatigue solely through signal processing or control strategies, this study demonstrates a wearable physical solution with measurable benefits at both muscular and perceptual levels. Future iterations could enhance functionality by extending assistance to additional joints and refining activation thresholds to better align with user intent and biomechanical demands [35].

The use of a bypass socket with non-disabled participants in prosthesis-related studies has been shown to reproduce compensatory movements similar to those observed in individuals with upper-limb amputations, making it useful for evaluating prosthetic devices in early design stages [22], [45]. However, this approach has limitations. First, the interface between the bypass socket and the limb of an non-disabled participant does not fully replicate the mechanical and sensory conditions of a residual limb, including soft tissue compliance, limb volume variability, and skin-prosthesis interactions [22]. Second, load distribution and prosthesis alignment differ from those of a custom-fitted socket, potentially affecting movement biomechanics and overall user experience [46]. Third, the absence of amputation-related neuromuscular adaptations, such as muscle atrophy, altered motor control, and proprioceptive deficits, may lead to differences in device use and perception. Accordingly, while current findings provide valuable preliminary insights, they may not fully represent performance or user experience with custom-fitted prostheses.

## VI. Conclusion

This study demonstrated that integrating a soft exoskeleton with a myoelectric prosthesis can significantly reduce muscle fatigue and perceived exertion during upper-limb tasks. Physiological and subjective data indicated decreased muscular demands, particularly in lifting tasks, with notable reductions in RMS activity in the deltoid, biceps, and lateral triceps, muscles commonly overloaded by prosthesis weight. Subjective measures, including RPE, SUS, and NASA-TLX scores, confirmed improved user experience alongside physical relief.

Limitations were observed in tasks requiring movements beyond the exoskeleton's support range, such as overhead reaching, highlighting mechanical constraints in the current design. Reduced statistical power in non-lifting tasks, due to fewer valid activations, further underscores the need for more robust sampling in future studies.

As a preliminary evaluation, this study focused on short-duration tasks to capture the immediate effects of exoskeleton assistance, acknowledging that muscle fatigue typically emerges after approximately five minutes of continuous use. The primary aim was to assess the feasibility of combining the ExoProsthesis by analyzing the muscles most affected during prosthesis use and determining whether the exoskeleton could reduce their fatigue. Future studies will extend the current findings by incorporating longer-duration, continuous cyclical tasks to evaluate sustained comfort, wear tolerance, and fatigue evolution over multiple activation cycles. These studies will also include portable indirect calorimetry to quantify whole-body metabolic cost and trunk muscle sEMG (erector spinae, multifidus, external oblique) with MVC normalization and RMS analysis. This will allow assessment of lumbar load redistribution and postural adaptations introduced by the exoskeleton's backpack-style force transmission. Additionally, future work will explore the integration of shoulder assistance and adaptive control strategies to enhance usability during overhead and high-load tasks.

Enhancing mechanical design, particularly by extending shoulder assistance and refining adaptive control mechanisms, could improve performance and intuitiveness. Developing a unilateral exoskeleton tailored to support the prosthetic limb may reduce bulk and weight, further enhancing wearability and task performance.

This study was conducted with non-disabled participants to verify the functional feasibility of combining an exoskeleton with a myoelectric prosthesis before involving upper-limb amputees. Testing the concept in a controlled setting is a necessary step; it would be impractical and ethically questionable to conduct trials directly with the target population without first ensuring that the hybrid system provides tangible benefits. Future studies should therefore include upper-limb amputees and real-world tasks to validate efficacy and usability. Long-term evaluations will clarify the exoskeleton's impact on fatigue, compensatory movements, and user acceptance, enabling more versatile and personalized systems suited to the complex demands of daily prosthesis use.

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